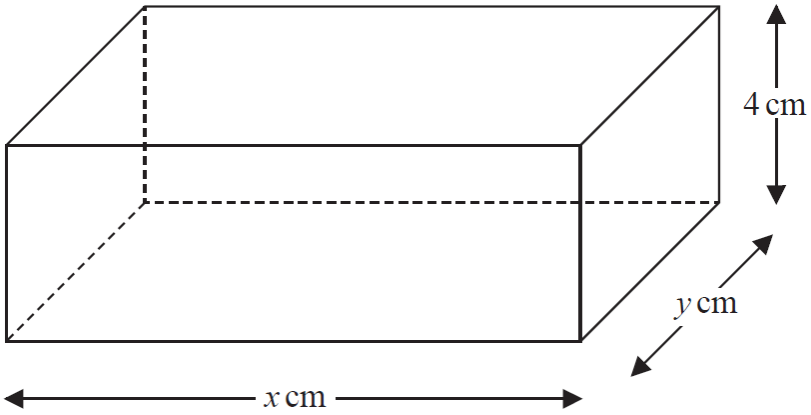
21 Here is a solid cuboid.



The volume of the cuboid is 119 cm³
The total surface area of the cuboid is 155.5 cm²

Given that $x > y$, work out the value of x and the value of y.
You must show all your working.

Volume: $4xy = 119$, so $xy = 29.75$.

1 mark
Volume equation.

Surface area: $2xy + 8x + 8y = 155.5$

1 mark
All six faces.

$59.5 + 8(x + y) = 155.5$, so $x + y = 12$.

$y = 12 - x$; $x(12 - x) = 29.75$

1 mark
One unknown.

$4x^2 - 48x + 119 = 0$

1 mark
Quadratic equal to 0.

$(2x - 17)(2x - 7) = 0$

$x = 8.5$ or 3.5; since $x > y$,

$x = 8.5$ cm and $y = 3.5$ cm.

1 mark
x = 8.5, y = 3.5.

$x =$ .....

$y =$ .....

(Total for Question 21 is 5 marks)